

**Math 271: Mathematical Methods**  
**Class Test 3 — Marking Scheme**

**Question 1: Prove that every sequence has a monotone subsequence. [50 Marks]**

This question assesses the student's ability to construct a formal proof for the **Monotone Subsequence Theorem**. The valid approaches I recognise here are the “peaks” method, the “direct construction” method, and an algorithmic presentation. Each is worth the full 50 marks.

**Approach 1: The ‘Peaks’ Method (Standard Proof)**

This is the most common and rigorous approach.

| Step                                       | Marks | Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Definition of a ‘Peak’                  | 10    | I award full marks for a precise definition. For example: “A term $x_p$ is a peak if $x_p \geq x_n$ for all $n > p$ .” I give partial credit (5–8 marks) for an intuitive but imprecise and informal definition, like calling it a “dominant element” or a term that is “larger than everything after it”, without stating the mathematical representation. A definition that is positively wrong (for instance, one quantifying over all $n$ , or over $n < p$ ) scores 0 for this item, and I then apply the rest of the rubric to whatever the student does with it. |
| 2. Stating the Two Exhaustive Cases        | 05    | I award these marks for clearly identifying that the proof splits into two mutually exclusive and exhaustive cases: (A) The sequence has infinitely many peaks, and (B) The sequence has finitely many peaks. I give partial credit (3 marks) for that split stated imprecisely, and nothing at all where the two cases on the page are not this split, are not exhaustive, or are false.                                                                                                                                                                               |
| 3. Proof for Case A: Infinitely Many Peaks | 15    | I award these marks for the argument that if there are infinitely many peaks, they form a non-increasing subsequence. The logic I look for is: Let the peaks be at indices $p_1 < p_2 < p_3 < \dots$ . By the definition of a peak, $x_{p_1} \geq x_{p_2} \geq x_{p_3} \geq \dots$ , which is a monotone subsequence. I award 10 of the 15 if the chain $x_{p_1} \geq x_{p_2} \geq \dots$ is asserted without deriving $x_{p_j} \geq x_{p_{j+1}}$ from the definition of a peak at $p_j$ together with $p_{j+1} > p_j$ .                                                |

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| 4. Proof for Case B:<br>Finitely Many Peaks | 20 | <p>I award these marks for the constructive proof of a strictly increasing subsequence.</p> <p>(4 marks) for correctly identifying the starting point, i.e. choosing an index <math>n_1</math> larger than every peak index. This is possible precisely because there are only finitely many peaks; if there is no peak at all, any <math>n_1</math> will do. I award 2 of the 4 for the phrasing “an index after the last peak”, which presupposes that a peak exists (it might not, for instance, <math>\{x_n\}_{n \in \mathbb{N}}</math> with <math>x_n = n</math>).</p> <p>(4 marks) for the observation that makes the induction work: since <math>n_1</math> exceeds every peak index, <i>no</i> index <math>n \geq n_1</math> is a peak.</p> <p>(12 marks) for the rigorous inductive construction: Since <math>x_{n_1}</math> is not a peak, there exists an <math>n_2 &gt; n_1</math> with <math>x_{n_2} &gt; x_{n_1}</math>. Since <math>x_{n_2}</math> is also not a peak (by the previous item), there exists an <math>n_3 &gt; n_2</math> with <math>x_{n_3} &gt; x_{n_2}</math>, and so on. The step here is uniform: the general clause “given <math>n_k \geq n_1</math>, since <math>x_{n_k}</math> is not a peak there is an <math>n_{k+1} &gt; n_k</math> with <math>x_{n_{k+1}} &gt; x_{n_k}</math>” is the first step verbatim with <math>n_1</math> replaced by <math>n_k</math>, and the preceding 4-mark item licenses it. I award the full 12 for a correct first step whose repetition is signalled by “and so on” (or “continuing inductively”, or similar), whether or not that general clause is written out. .</p> |
| Deductions                                  | —  | <p>(−2 to −8 marks) for imprecise notation (e.g. using <math>x_2, x_3</math> for subsequence terms instead of <math>x_{n_k}</math>) or lack of overall clarity.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

## Approach 2: Direct Construction / Dichotomy Method

Many students produce the following write-up without ever naming what a peak is. It is a legitimate alternative way of writing the *same* proof: the “index whose value is never matched later” is a peak under another name. I have therefore itemised it exactly as Approach 1 — an item for the dominating-index notion (whether or not it is given a name), an item for the two exhaustive cases, and one item for each of the two constructions.

| Step                                | Marks | Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| 1. The dominating-index notion      | 10    | I award full marks for isolating, precisely, the notion the proof runs on, namely an index that dominates its own tail: “ $n$ is such an index if $x_k < x_n$ for every $k > n$ ”, or equivalently “there is no $k > n$ with $x_k \geq x_n$ ”. Equivalently, the set $S_n = \{x_k : k > n \text{ and } x_k \geq x_n\}$ is empty. 5–8 marks for an informal version, e.g. “we can always find a larger term later, or this eventually fails”, without the quantified statement.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 2. Stating the Two Exhaustive Cases | 05    | I award these marks for stating the two cases in a mutually exclusive and exhaustive form: (A) for every $n$ there is a $k > n$ with $x_k \geq x_n$ (the search for a later term that is at least as large never fails); (B) there is an $N$ such that no $k > N$ satisfies $x_k \geq x_N$ (the search fails at $N$ ). I give partial credit (3 marks) for that split stated imprecisely, and nothing at all where the two cases on the page are not this split, are not exhaustive, or are false. <i>Note:</i> the dichotomy “ $S_n$ is finite for all $n$ , or infinite for some $n$ ” is <i>not</i> the case split used in the two constructions below <i>The two cases must be applied to tails</i> , see item 4.                                                                                                                                                                                                                                          |
| 3. Case A: the search never fails   | 15    | I award these marks for the inductive construction of a non-decreasing subsequence. Logic: Choose $n_1$ . Since we can always find a later term that is at least as large, choose $n_2 > n_1$ with $x_{n_2} \geq x_{n_1}$ . Then choose $n_3 > n_2$ with $x_{n_3} \geq x_{n_2}$ , and so on. The step is uniform — the Case A hypothesis is stated for <i>every</i> $n$ , so the clause used at stage $k$ is the first one with $n_1$ replaced by $n_k$ — and the recursion may be signalled by “and so on” and still earn the full 15. I award 10 of the 15 only where the uniformity is missing, i.e. where the hypothesis has been recorded at the starting index alone, so that nothing in the answer script licenses the subsequent applications. Note that the hypothesis gives $x_{n_{k+1}} \geq x_{n_k}$ , so the subsequence obtained is non-decreasing, not necessarily increasing; an answer script that claims strict increase here loses 2 marks. |

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| 4. Case B: the search fails somewhere | 20 | <p>I award these marks for showing how the failure in Case A produces a non-increasing subsequence.</p> <p>(4 marks) for the setup: assume there is some <math>n_1</math> such that there is no <math>k &gt; n_1</math> with <math>x_k \geq x_{n_1}</math>; this implies <math>x_k &lt; x_{n_1}</math> for all <math>k &gt; n_1</math>.</p> <p>(4 marks) for recognising that the dichotomy must now be re-applied <i>to the tail after <math>n_1</math></i>, and not merely repeated: the hypothesis of Case B supplies only <i>one</i> such index, and there are sequences possessing exactly one (for example <math>x_1 = 2</math>, <math>x_n = 1 - \frac{1}{n}</math> for <math>n \geq 2</math>). I do not award these 4 marks for the bare assertion “a similar logic can be applied”, with <math>n_2 = n_1 + 1</math> and no further argument, and it does not earn the 12 below either.</p> <p>(12 marks) for the construction proper: apply the dichotomy to the tail beyond <math>n_1</math>. If that tail again contains an index dominating its own tail, take it to be <math>n_2</math>, then repeat, and the resulting <math>x_{n_1} &gt; x_{n_2} &gt; x_{n_3} &gt; \dots</math> is a strictly decreasing subsequence (the repetition here is legitimate because the dichotomy is re-applied to a new tail at each stage, which is the content of the preceding item; it is <i>not</i> the case hypothesis being repeated, and that is why the bare “and so on” fails here — this is the non-uniform application; if at some stage the tail contains no such index, then inside that tail every term is followed by a term at least as large, and item 3’s construction runs there and yields a non-decreasing subsequence. I award 6 of the 12 for one of the two branches carried out correctly.</p> |
| Deductions                            | —  | <p>(−2 to −8 marks) for imprecise notation (e.g. using <math>x_2, x_3</math> for subsequence terms instead of <math>x_{n_k}</math>) or lack of overall clarity, subject to C4.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

### Approach 3: The Algorithmic Method

A student may present the proof as an algorithm that reads the sequence and outputs the indices of a monotone subsequence. This is a legitimate write-up provided the two things that make an algorithm a proof are supplied: (i) every instruction is executable, i.e. each search that the algorithm performs is guaranteed to succeed, and (ii) an invariant is exhibited showing that what the algorithm outputs has the required property. I mark the mathematics below, not the notation: pseudocode, numbered English instructions, or a flow chart are all acceptable.

The intended algorithm, in the notation of Approach 1 (an index  $p$  is a *peak* if  $x_p \geq x_k$  for all  $k > p$ ; the student may call it a dominating index or nothing at all):

**Algorithm MonotoneSubsequence****Input:** a sequence  $x_1, x_2, x_3, \dots$  of real numbers.**Output:** indices  $n_1 < n_2 < n_3 < \dots$  such that  $x_{n_1}, x_{n_2}, x_{n_3}, \dots$  is monotone.

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1.  Test whether the set  $P = \{p : x_p \geq x_k \text{ for all } k > p\}$  of peaks is infinite.
2.  if  $P$  is infinite then    // Peak branch
3.       $n_1 := \min P$ 
4.      for  $j = 1, 2, 3, \dots$  do
5.           $n_{j+1} := \min\{p \in P : p > n_j\}$ 
6.          output  $n_1 < n_2 < n_3 < \dots$     // values non-increasing
7.  else    // Climb branch
8.       $N := 1 + \max P$     (and  $N := 1$  if  $P = \emptyset$ )
9.       $n_1 := N$ 
10.     for  $j = 1, 2, 3, \dots$  do
11.          $n_{j+1} := \min\{k > n_j : x_k > x_{n_j}\}$ 
12.         output  $n_1 < n_2 < n_3 < \dots$     // values strictly increasing

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In English: first ask whether peaks keep occurring forever. If they do, simply list them in order; each one dominates everything after it, in particular the next one, so the values read off are non-increasing. If they do not, move past the last peak and start climbing: from any position beyond the last peak the current term is not a peak, so some later term is strictly larger; jump to the first such term and repeat. The climb can never get stuck, because every position it reaches is again beyond the last peak. Taking the *least* admissible index at each step is not needed for correctness; it only makes the output uniquely determined.

| Step                              | Marks | Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Specification                  | 06    | I award these marks for stating precisely what the algorithm is given and what it must produce: an infinite list of <i>strictly increasing</i> indices whose terms are non-decreasing or non-increasing. I award 3 if the output is described only as “a monotone subsequence” without the requirement on the indices.                                                                                                                               |
| 2. The Test Driving the Branch    | 10    | I award these marks for the test on which the algorithm branches, stated precisely: whether the peaks (or dominating indices, however named) are infinite in number. I give partial credit (5–8 marks) for an informal test such as “check whether the process of finding larger terms can go on forever” without the mathematical representation. This item is the counterpart of item 1 of Approaches 1 and 2, and I mark it to the same standard. |
| 3. Exhaustiveness of the Branches | 04    | I award these marks for observing that the test’s two outcomes are mutually exclusive and exhaustive, so that the algorithm always enters exactly one branch. Where the two outcomes on the page are not the branch test, are not exhaustive, or are false, the item scores 0.                                                                                                                                                                       |

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|-----------------|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4. Peak Branch  | 12 | <p>(4 marks) for the loop being well defined: because <math>P</math> is infinite, <math>\{p \in P : p &gt; n_j\}</math> is non-empty for every <math>j</math>, so line 5 always finds a value; the outputs strictly increase by construction.</p> <p>(8 marks) for the correctness argument: <math>n_j \in P</math> and <math>n_{j+1} &gt; n_j</math>, so by the definition of a peak <math>x_{n_{j+1}} \leq x_{n_j}</math>, and the output values are non-increasing.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 5. Climb Branch | 18 | <p>(4 marks) for line 8: <math>P</math> is finite, so it has a largest element and <math>N</math> is well defined; if <math>P = \emptyset</math> any starting index serves.</p> <p>(8 marks) for the crucial guarantee that line 11 never fails: every index <math>n \geq N</math> lies beyond the last peak, hence is not a peak, hence has some <math>k &gt; n</math> with <math>x_k &gt; x_n</math>; so the set searched in line 11 is non-empty. An answer script that runs the climb without justifying that the search always succeeds earns 0 of these 8 marks: this guarantee is the fact that makes the loop uniform, and it is what an “and so on” has to be licensed on. I charge it here, once, and not again against the invariant below.</p> <p>(6 marks) for the loop invariant: <math>n_j \geq N</math> and <math>n_{j+1} &gt; n_j</math> for all <math>j</math> (so the previous item applies at every step), and <math>x_{n_{j+1}} &gt; x_{n_j}</math>, whence the output values are strictly increasing.</p> |
| Deductions      | —  | <p>(−2 to −8 marks) for imprecise notation or lack of overall clarity. In particular, an algorithm whose loop is written as “repeat until done” or which is presented as <i>terminating</i> (it does not: it produces an infinite output, and the correctness argument is the invariant, not a halting condition) attracts a deduction in this range, not a loss of the items above.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

**Note.** The test in line 1 is not something a computing device could carry out — deciding whether an index is a peak involves infinitely many comparisons — and I do not penalise students for the algorithm being non-effective in this sense; nor do I reward them for claiming that it is effective. The algorithm is a way of *presenting* the construction; what I mark is the construction. A student who remarks on the point earns no extra marks, and no deduction either.

## Question 2: Prove that any Cauchy sequence of real numbers converges. [50 Marks]

This question assesses the student’s ability to synthesize given hints and fundamental theorems. The two primary valid approaches use either the Bolzano–Weierstrass Theorem or the result from Question 1. Both are worth the full 50 marks; note that the paper’s own hints point towards Approach 2, and I do not penalise an answer script for not naming the Bolzano–Weierstrass Theorem if it supplies the same content by another route.

**Approach 1: Using the Bolzano–Weierstrass Theorem**

| Step                                         | Marks | Notes                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------------------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. State the Definition of Cauchy Sequence   | 04    | I award these marks for formally defining $\{x_n\}$ a Cauchy sequence. The operative word is <i>formally</i> : for every $\varepsilon > 0$ there is an integer $N$ such that $ x_i - x_j  < \varepsilon$ for all $i, j \geq N$ , with the quantifiers in this order and $\varepsilon$ arbitrary. I award 2 if the definition is stated with the quantifiers in the wrong order or with $N$ independent of $\varepsilon$ . |
| 2. Using Hint 1: Boundedness                 | 04    | I award these marks for formally stating that since $\{x_n\}$ is a Cauchy sequence, it must be bounded: there is $B > 0$ with $ x_n  \leq B$ for all $n$ . Again the operative word is <i>formally</i> . I award 2 for an unquantified assertion that the sequence “does not run away”.                                                                                                                                   |
| 3. Invoking the Bolzano–Weierstrass Theorem  | 22    | This is the cornerstone of this proof. I award full marks for stating that since the sequence is bounded, it lies in a compact set - a closed interval and by the Bolzano–Weierstrass Theorem, it must have a convergent subsequence, with the hypothesis of the theorem checked. I award 16 if the theorem is applied correctly but its hypothesis is not checked.                                                       |
| 4. Using Hint 2: Convergence of the Sequence | 20    | I award these marks for the final logical step: stating that since $\{x_n\}$ is a Cauchy sequence and it has a convergent subsequence (from step 3), the entire sequence $\{x_n\}$ converges to the same limit, citing the hint. I award 12 if the hint is used but the limit of the whole sequence is not identified with the limit of the subsequence.                                                                  |
| Deductions                                   | —     | (−2 to −8 marks) for lack of clarity, poor structure, or minor imprecisions in terminology.                                                                                                                                                                                                                                                                                                                               |

**Approach 2: Using the Monotone Subsequence Theorem (Result from Q1)**

This is a valid alternative strategy that many students successfully identified.

| Step                                       | Marks | Notes                     |
|--------------------------------------------|-------|---------------------------|
| 1. State the Definition of Cauchy Sequence | 04    | As in Approach 1, item 1. |
| 2. Using Hint 1: Boundedness               | 04    | As in Approach 1, item 2. |

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|---------------------------------------------------|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3. Invoking the Monotone Subsequence Theorem (Q1) | 06 | I award these marks for correctly applying the result from Question 1: the sequence has a monotone subsequence $\{x_{n_k}\}$ . Note that Question 1 needs <i>no</i> hypothesis: boundedness is not what produces the monotone subsequence, and is used only at item 4. An answer script that writes “since the sequence is bounded, it must contain a monotone subsequence” still earns these 6 marks, but attracts a $-2$ deduction for the spurious dependency.                                                                                                                                                                   |
| 4. Invoking the Monotone Convergence Theorem      | 16 | I award these marks for identifying the key insight: the subsequence $\{x_{n_k}\}$ is both monotone (from step 3) and bounded (since the parent sequence is bounded, and $ x_{n_k}  \leq B$ for every $k$ ). Therefore, by the Monotone Convergence Theorem, the subsequence $\{x_{n_k}\}$ converges.<br>(6 marks) for noting that a subsequence of a bounded sequence is bounded; (10 marks) for the appeal to the Monotone Convergence Theorem with both of its hypotheses in place. An answer script that handles only the non-decreasing case, without remarking that the non-increasing case is symmetric, earns 10 of the 16. |
| 5. Using Hint 2: Convergence of the Sequence      | 20 | I award these marks for the final logical step: stating that since $\{x_n\}$ is a Cauchy sequence and it has a convergent subsequence (from step 4), the entire sequence $\{x_n\}$ converges to the same limit, citing the hint. I award 12 if the hint is used but the limit of the whole sequence is not identified with the limit of the subsequence.                                                                                                                                                                                                                                                                            |
| Deductions                                        | —  | ( $-2$ to $-8$ marks) for errors.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |